UI-MOPD: Multi-Platform On-Policy Distillation for Unified GUI Agents

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Paper version: arXiv 2607.04425v2 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Naively mixing supervision or merging specialized models can blur native behaviors and produce imbalanced performance.*

Desktop and mobile share **grounding** but clash in **action semantics**.

Why one policy is hard

1. **Shared but conflicting.** “Return”: close window vs back button.
2. **Mixing damages.** Desktop-only SFT: MobileWorld 7.7% → 0%.
3. **Teachers already know.** Each anchors its own platform.

Evaluation setup

OSWorld · 361 tasks | MobileWorld · 117 tasks | Qwen3-VL-8B student | 32B platform teachers | Uni-GUI ~10K trajs | 64 × H100

- **Online:** OSWorld 361 + MobileWorld 117 tasks; success rate.
- **Offline (controls only):** AndroidControl⋆, ScreenSpot-Pro/V2, OSWorld-G.

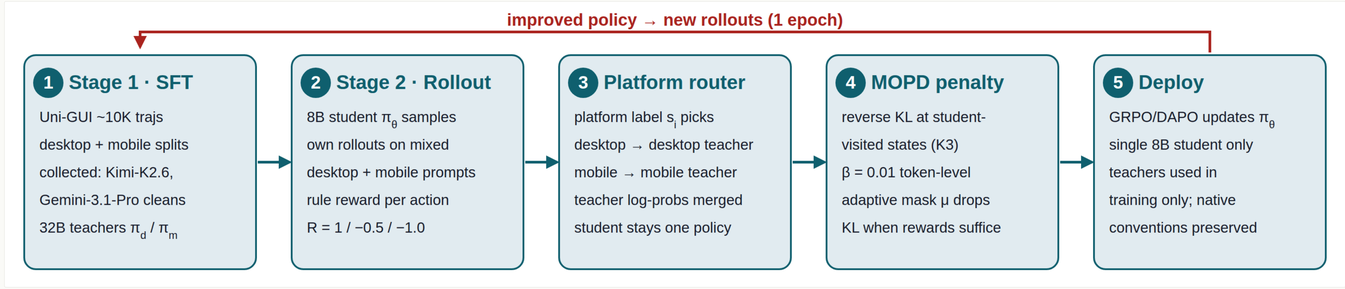
Our read (CUA Radar)

64	★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; matched 8B baselines; ablation; compute. **Not reported:** seeds/variance; held-out envs; code check.
- **Flags:** none (AP7: no).
- Pattern R01. Authors’ numbers, not rerun.

Method: route each rollout to its platform teacher

- **Sees / acts:** screenshots in; JSON tool calls out — computer_use (desktop) or mobile_use (mobile).
- **Teachers:** SFT two Qwen3-VL-32B specialists on Uni-GUI (~10K trajectories, ~160K steps).
- **Signal:** rule reward 1 / −0.5 / −1.0 per action + reverse-KL anchor from the routed teacher (β = 0.01).



Our redraw (not the paper’s figure): the two-stage loop with platform-routed teachers.

$$\hat{D}_{\mathrm{KL}}^{(t)} = \rho_t - \delta_t - 1, \qquad \delta_t = \log \pi_{\mathrm{ref}}(y_t \mid h_t) - \log \pi_\theta(y_t \mid h_t)$$

K3 token-level KL to the routed teacher (Eq. 5)

Results: online success rate (%)

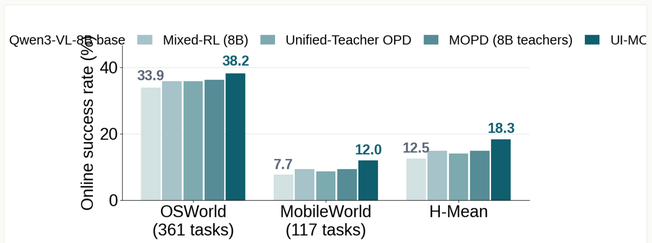


Table 1, redrawn: 8B student; OSWorld 361 tasks, MobileWorld 117.

H-Mean 18.3% matches Qwen3-VL-235B-A22B-Thinking (18.3%)

SO WHAT → *H-Mean is the paper’s balanced metric (Sec. 4.2). No error bars: the paper reports no seeds or variance.*

Ablation: MinGain over the 8B base

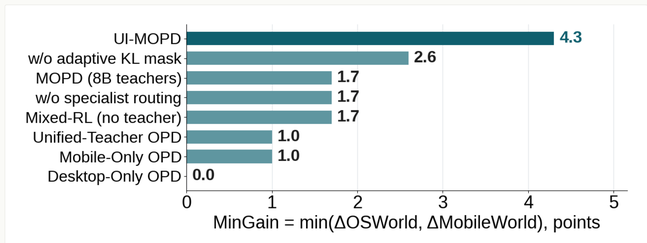


Table 2, redrawn: same 8B student and Stage-2 data; “w/o routing” picks teachers at random.

SO WHAT → *Routing, specialist teachers and the adaptive mask each contribute. 8B teachers (1.7) and a unified 32B teacher (1.0) both fall short: specialization and scale both matter. Only full UI-MOPD gains 4.3 points on both platforms. The mask matters because imitating already-rewarded groups restricts exploration (Sec. 3.2).*

Headline Numbers

38.2%
OSWorld success rate
vs 33.9% for the 8B base

- **12.0%** MobileWorld
- **18.3%** H-Mean
- **+4.3** MinGain pts

The paper’s own controls

CONTROL	WHAT VARIES	OSW / MW
Mixed-SFT (8B)	mixed data	35.0% / 6.4%
Seq-RL (D→M)	training order	33.7% / 11.1%
TIES merge	static merge	36.8% / 0%
Desktop-Only OPD	one 32B teacher	36.0% / 7.7%
Offline retention	SS-Pro/V2, OSW-G	within 0.6 pts

Every matched 8B baseline trails on ≥1 platform.

Takeaway

One 8B policy, two teachers as anchors: 38.2% OSWorld, 12.0% MobileWorld, grounding within 0.6 pts.

~10K Uni-GUI trajs | teachers: train-time only

Scan to Read

Paper | Code